

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
Faulty RAM	PC does not boot, continuous beep sounds, or frequent blue screen errors.
Failed Power Supply (SMPS)	Computer does not turn on or shuts down unexpectedly
Hard Disk/SSD Failure	Slow performance, clicking noise, or operating system fails to load.
Overheating CPU	Random shutdowns, automatic restarts, or high fan speed.
Faulty Graphics Card (GPU)	No display, screen artifacts, or distorted graphics on the monitor.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

1. **Check the power outlet**

- Plug another device into the same socket to ensure power is available.

2. **Inspect the charger**

- Look for cuts, bends, or damage in the cable and adapter.

3. **Check the charging port**

- Inspect the laptop's charging port for dust, bent pins, or loose connections.

4. **Test the charger with a multimeter**

- Measure the output voltage and compare it with the voltage printed on the charger label.

5. Check the battery connection (if removable)

- Remove and reinsert the battery to ensure it is properly connected.

6. Try another compatible charger

- If available, test with a known working charger to identify whether the charger is faulty.

7. Observe the charging LED

- Check whether the charging indicator lights up or blinks, which may indicate power or battery issues.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Item	Result
Expected Voltage	4.75V – 5.25V DC
Measured Voltage	5.08V DC (<i>sample reading from a virtual simulator</i>)
Status	Within the expected USB voltage range

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint:
Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Steps

1. Turn the multimeter to **DC Voltage (20V range or auto-range)**.
2. Disconnect the ATX connector from the motherboard.
3. Turn on the power supply (using the paperclip method if required).
4. Place the **black probe** on any **black wire (Ground)**.
5. Place the **red probe** on the **yellow wire (+12V)**.
6. Read the voltage displayed on the multimeter.